

Evaluating the 2019 Child Marriage Ban in Brazil

Mechanisms, a Marriage Market Model, and Identification Strategy

Research Team

Research Design Note for Coauthor Discussion

April 2026

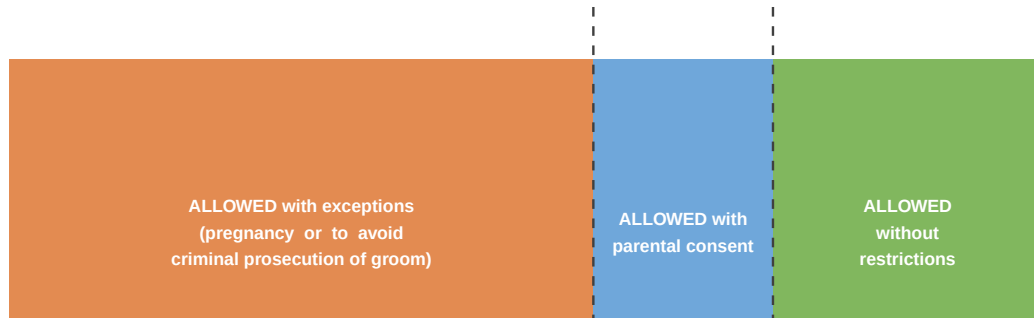
Motivation

Before Lei 13.811/2019: a permissive regime below age 16

Art. 1.520 of the Civil Code allowed a judge to grant marriage *below age 16* in two circumstances: (i) pregnancy, or (ii) to avoid criminal prosecution of the groom. No minimum age floor existed — exceptions were granted case by case.

Legal regime for marriage — Before Lei 13.811/2019

Art. 1.520 of the Civil Code: below age 16, exceptions were granted by a judge



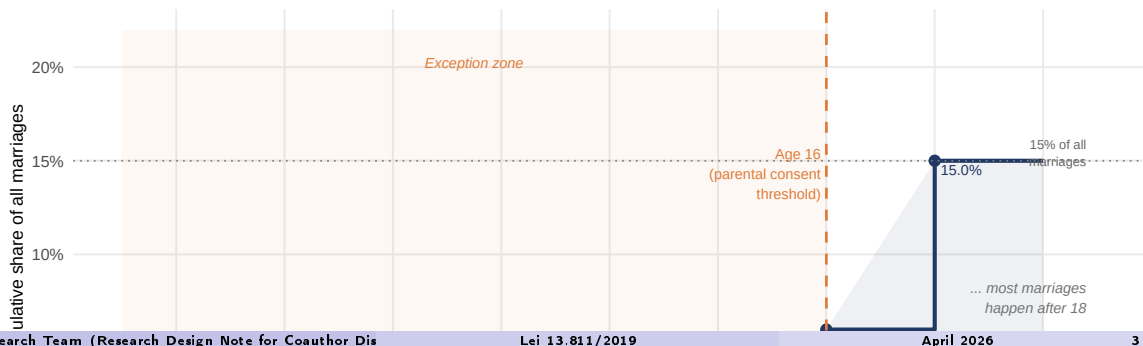
Motivation

Baseline distribution of formal marriages before 2019

Under this regime, ~85% of formal child marriages are concentrated at ages 16–17 (parental-consent zone), but the exception clause generates a **small tail below age 16**. This tail is our pre-reform baseline — it should vanish after 2019 *if* the ban is enforced.

Expected CDF of formal child marriages — Before Lei 13.811/2019

Y-axis: fraction of ALL marriages (including post-18); small exception-zone tail visible below age 16



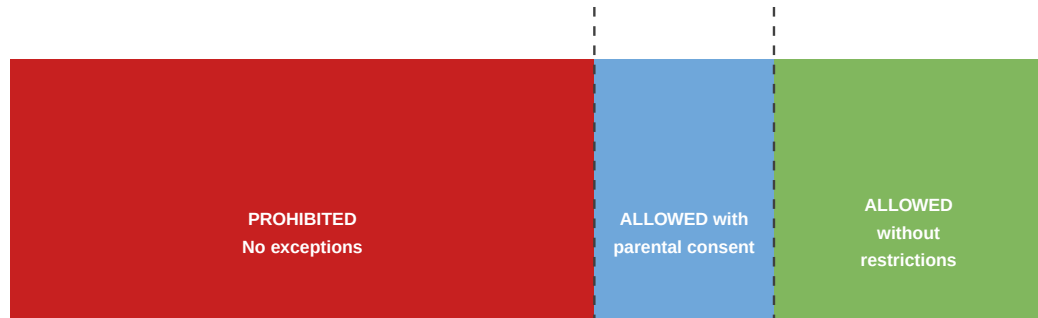
Motivation

Lei 13.811/2019: an unconditional prohibition below age 16

Enacted 12 March 2019, the law eliminated *all* exceptions — formal marriage below age 16 is now **absolutely prohibited** under any circumstance. This creates a **clean, well-defined treatment threshold**: a hard age floor with no judicial discretion.

Legal regime for marriage — After Lei 13.811/2019

Lei 13.811/2019: formal marriage below age 16 is absolutely prohibited



Motivation

Hypothesis 1 — Formal marriages below 16 should drop to zero

Under full enforcement, the Registro Civil should show **no** formal marriages below age 16 after 2019. We test this with an event study on the formal-marriage hazard and a McCrary bunching test at the legal threshold. Any residual mass below 16 in administrative data signals **non-compliance or evasion**.

Expected CDF of formal child marriages — After Lei 13.811/2019

Under full enforcement: zero mass below 16; plateau slightly lower than pre-2019



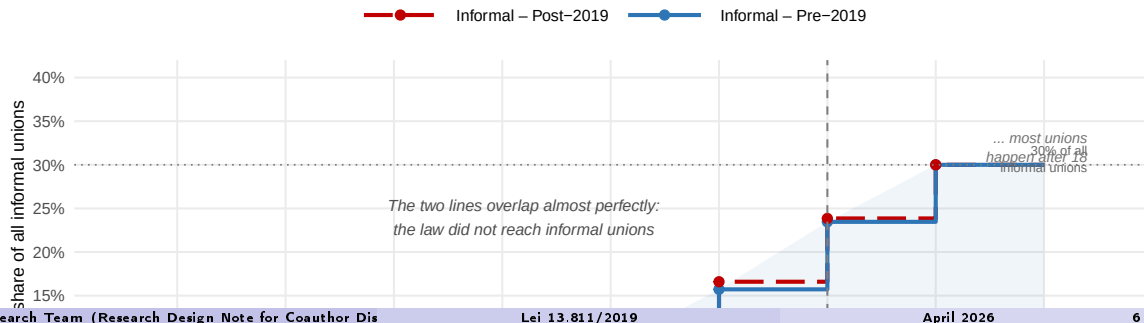
Motivation

Hypothesis 2 — Informal unions are outside the law's reach

The civil-registry ban covers *only* formal marriages. Informal unions — **89.5% of child unions** (Census 2010) — are legally unreachable. If families substitute from formal to informal channels, the informal CDF should be **unchanged or larger** after 2019. Divergence between the two margins identifies **real reduction** vs. **displacement**.

Expected CDF of informal child unions — Before vs. After 2019

Informal unions not covered by civil-registry law — distribution nearly unchanged



Outline

- 1 Motivation and Research Question
- 2 Child Marriage in Brazil: Stylized Facts
- 3 Literature: Global vs. Brazilian
- 4 Mechanisms: A 2×2 Typology
- 5 A Model of the Brazilian Marriage Market
- 6 Empirical Strategy
- 7 Data Sources and Variable Construction
- 8 Next Steps
- 9 Structural Estimation

- Brazil ranks **fourth worldwide** in absolute number of women married as children (Margarido 2024).
- **Lei 13.811/2019** eliminated all exceptions to the minimum marriage age of 16 — previously, pregnancy or avoidance of criminal prosecution permitted marriage below 16.
- Yet formal and informal child unions persist: Registro Civil 2021 records **45.4 girls married per day** formally (Margarido 2024).
- Global evidence on minimum-age laws is mixed: some bind (Bharadwaj 2015), many fail to enforce (Collin 2016); administrative data systematically overstates law effects (Blank 2020).

Did the 2019 ban actually reduce child marriage in Brazil — and which marriages persisted?

Research Questions

Primary question

Did Lei 13.811/2019 causally reduce child marriage in Brazil, and through which margins (formal vs. informal, age, region)?

Secondary questions

- 1 What **mechanisms** sustain child marriage in Brazil — economic pressure, normative pressure, or both?
- 2 Which marriages are **elastic to income policy** (e.g., Bolsa Família) vs. **inelastic** (norm-driven, enforcement-proof)?
- 3 Can we quantify a **“norm-irreducible” share** of child marriages that legal and economic interventions alone cannot reach?

Contribution

First causal evaluation of Lei 13.811/2019 combining a marriage-market model, heterogeneous-mechanism estimation, and formal/informal margin comparison.

Before 12 March 2019 (Art. 1.520, CC):

- Minimum marriage age: 16 with parental consent; 18 without.
- **Exceptions below 16:** pregnancy or to avoid criminal prosecution of the groom.

After Lei 13.811/2019:

- Absolute minimum: 16 (with parental consent).
- **No exceptions.**
- Informal unions remain legally unregulated.

Key enforcement concern

The law only affects *formal* marriages. Informal unions, which constitute **89.5% of child unions** in the 2010 Census (Margarido 2024), fall outside civil registry jurisdiction.

Stylized Facts (I): Prevalence and Demographics

- CARDOSO ET AL. 2022 (PNS 2013): **3.9% prevalence** of union before 18 among under-18s; overwhelmingly female, Black/brown, lower-income, Northern/Northeastern.
- MARGARIDO 2024 (Census 2010): **655,935 children** in some form of union; girls outnumber boys 5:1; **89.5% informal**.
- PINTO ET AL. 2024 (SINASC 2011–2021): **127,022 live births** to mothers aged 10–14; **21.1%** declared married or in stable union. Moran's $I = 0.47$ — strong spatial clustering in North and Center-West.
- AGUIAR ET AL. 2025 (SINAN 2014–2021): **3,681 cases** of marital rape of girls ≤ 14 ; 61% recurrent; concentrated in Acre, Pernambuco, Pará, Amazonas.

Stylized Facts (II): Who, Where, Why

Who

- Girls, ages 12–17 (mode 14–15)
- Black/brown (65–75% across studies)
- Lower household education and income
- Often pregnant at time of union formation

Where

- Rural North and Northeast
- Center-West emerging (Pinto 2024)
- Peripheries of major cities (Margarido 2021)

Why (qualitative evidence)

- Control of female sexuality / family honor (Taylor & Fonseca 2015)
- Poverty / “exit option” (Veiga & Zanello 2020)
- Pregnancy “repair” union (Taylor & Fonseca 2015)
- Normalization via social media (Huhn 2022)
- Intergenerational transmission (Asadullah 2018, globally)

Stylized Facts (III): What the PBF Evidence Already Tells Us

VASCONCELOS & GRIEBELER (2017, 2023) use Propensity Score Matching on PNADC data:

- Bolsa Família **reduces child marriage** among poor and extremely poor girls.
- **Protective effect is concentrated at ages 16–18.**
- **Null effect at 14–15:** income transfers do not move the youngest unions.

Interpretation

The age-heterogeneity in PBF response is already **prima facie evidence** that a large share of the youngest marriages are driven by mechanisms other than economic pressure — the same mechanisms a legal ban would need to reach.

Global Literature — Core Findings

Study	Context / Method	Key finding
Field & Ambrus (2008)	Bangladesh / menar- che IV	+1 yr marriage delay → +0.22 yr schooling
Dahl (2010)	USA / state law IV	Early marriage → +31 pp poverty for women
Bharadwaj (2015)	Mississippi 1957 / DiD	Law reform: −75% marriages, +3% enrollment
Duflo et al. (2015)	Kenya / RCT	Education subsidy beats HIV education for delay
Wahhaj (2015)	Bangladesh / OLG model	Reputational equilibrium sustains early marriage
Ashraf et al. (2016)	Indonesia / Zambia	School supply binds in bride-price groups only
Collin & Talbot (2016)	60+ countries	Most minimum-age laws poorly enforced
Kidman (2016)	34 countries / DHS	Child marriage → OR ≈ 1.5 for IPV
Chari et al. (2017)	India / menarche IV	Intergenerational: +3.1% child school enrollment
Corno et al. (2019)	SSA vs. India	Bride price ↑ vs. dowry ↓ under droughts
Blank et al. (2020)	USA / admin vs. sur- vey	Admin data overstates law effects
Chow & Vivalt (2022)	Ethiopia / DiD	Community interventions reduce marriage 4–7 pp
Wilson (2022)	LMICs / DiD	Age-18 bans raise marriage age, first-birth age, em- ployment

Brazilian Literature — Core Findings

Study	Data / Method	Key finding
Taylor & Fonseca (2015)	Pará/Maranhão / mixed	Honor, sexuality control, pregnancy trigger unions
Vasconcelos & Griebeler (2017)	PNADC / PSM	PBF protects girls in extreme poverty
Veiga & Zanello (2020)	Goiás / qualitative	“Choosing and being chosen”; constrained agency
Cargnin et al. (2021)	Rio Branco / SINAN	41.2% of 10–14 yr sexual-violence victims in union
Margarido (2021)	São Paulo / ethnography	Protection-network failures in metropolitan periphery
Cardoso et al. (2022)	PNS 2013 / Poisson	3.9% prevalence; intersectional determinants
Huhn (2022)	Digital ethnography	Social media normalizes the practice
Vasconcelos & Griebeler (2023)	PNADC / PSM	PBF effect concentrated at 16–18; null at 14–15
Margarido (2024)	Census + Registro Civil	655k unions (Census); 45.4 formal marriages/day (2021); agency typology
Pinto et al. (2024)	SINASC / spatial	Moran's I = 0.47; North/Center-West clusters
Aguiar et al. (2025)	SINAN / multivariate	3,681 marital rapes of girls ≤ 14 (2014–2021)

Contrasting the Two Literatures

Dimension	Global literature	Brazilian literature
Causal identification	Strong (IV, DiD, RCT)	Mostly descriptive / PSM
Economic mechanism	Bride price, dowry, income shocks	Poverty as exit option; PBF
Norm mechanism	Formal equilibrium models (Wahhaj)	Qualitative, ethnographic
Violence	IPV as outcome (Kidman)	Central frame (SINAN, SINASC)
Race/class	Rarely foregrounded	Intersectionality central
Agency	Girls as passive	Rich constrained-agency theory
Law enforcement	Quantitatively skeptical	Institutionally skeptical

Our opportunity

Import global causal-identification tools into the richer Brazilian descriptive, health-system, and intersectional framing — and close the methodological gap.

Mechanisms: the 2×2 Typology

	Low normative pressure	High normative pressure
High economic pressure	Type A: Pure poverty Income policy eliminates marriage. PBF bites.	Type B: Reinforced Income policy partially reduces.
Low economic pressure	Type D: Neither Marriage unlikely.	Type C: Pure norm Income policy has zero effect. Law may or may not bind.

- Vasconcelos & Griebeler's **null effect at ages 14–15** is consistent with a large share of Type C in that age window.
- A legal ban should, in principle, reach Types B and C — **if enforced**.
- Types B and C are also where post-pregnancy “repair” unions concentrate.

Which Marriages Persist Under an Income Policy?

Expected to persist under a pure income policy (PBF or universal transfer):

- 1 **Very young unions (< 15)**: age floor itself is normative; income elasticity ≈ 0 .
- 2 **Post-pregnancy “repair” unions**: causality runs pregnancy $\rightarrow$ marriage; transfers don't intervene in time.
- 3 **High-norm regions / families**: Northern interior, intergenerational transmission.
- 4 **Partner-coerced unions**: SINAN data (Aguiar 2025) shows perpetrator is often older, recurrent — income to parents may be inert or counterproductive.
- 5 **Informal unions below legal threshold**: evade both Registro Civil and formal policy targeting.

Testable implication for Lei 13.811/2019

If the 2019 ban reduces only formal marriages but leaves informal unions unchanged, it shifts margins without reducing total child marriage. This is the **Collin (2016) / Blank (2020) concern** applied to Brazil.

Modeling Choices for the Brazilian Context

- **No formal bride price or dowry** — rules out Corno (2019) sign-switching directly. Economic channel operates through consumption and labor.
- **Pregnancy is a major trigger** of union formation (Taylor & Fonseca 2015) — must enter the model as a state variable.
- **Informal $\gg$ formal** — agents face a margin choice (formal vs. informal union), modulated by law.
- **Agency is constrained but real** (Margarido 2024, Veiga & Zanello 2020) — decision is joint family-girl, not pure parental.
- **Norms vary intergenerationally** — usable as identification variation.

A Discrete-Choice Marriage Timing Model

Household i with daughter age $a \in \{10, 11, \dots, 17\}$. In each period decides: *marry formally* ($M = F$), *marry informally* ($M = I$), or *wait* ($M = W$).

Flow utility by choice:

$$\begin{aligned}U_i^F(a) &= W_{ia} + \kappa_i N_i + \mu_i R(a) - \phi \cdot \mathbb{I}[a < \bar{a}_t] + \varepsilon_{ia}^F \\U_i^I(a) &= W_{ia} + (\kappa_i - \lambda) N_i + \mu_i R(a) + \varepsilon_{ia}^I \\U_i^W(a) &= Y_i + s(a) + \delta E[V(a+1)] + \varepsilon_{ia}^W\end{aligned}$$

where

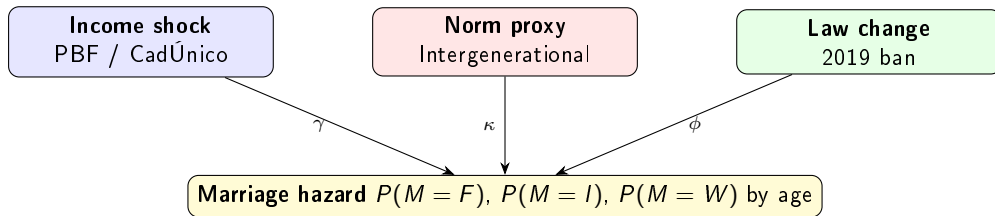
- W_{ia} : economic payoff (consumption relief, pregnancy resolution)
- N_i : household normative index; κ_i : weight on norms
- $R(a)$: reputational risk increasing in age (Wahhaj channel); μ_i : sensitivity
- ϕ : **legal-penalty term**, active only when age < minimum at time t
- $\lambda > 0$: norms slightly penalize informal relative to formal union
- $s(a)$: schooling return; Y_i : household income; δ : discount factor

Predictions from the Model

- 1 **Law effect on formal margin:** Lei 13.811/2019 raises ϕ from 0 (exception applicable) to large positive for $a < 16$. Formal marriages under 16 → **sharp drop** in Registro Civil.
- 2 **Substitution to informal margin:** ϕ only affects U^F , not U^I . If $\kappa_i N_i$ is large, family substitutes into informal union. Net effect on total unions: **ambiguous**.
- 3 **Heterogeneity by norm:**
 - Low N_i : law reduces marriages (little substitution, low baseline).
 - High N_i : law triggers substitution, minimal reduction in total unions.
- 4 **Age bunching:** McCrary-style discontinuity expected at 16 post-2019 in formal data; weaker in informal data.
- 5 **Pregnancy interaction:** Removal of pregnancy exception → sharpest compliance effect for pregnant under-16 cases.

Identifying the Parameters

The model is identified off three sources of variation:



- γ (income elasticity): PBF variation (PSM or eligibility RDD, as in Vasconcelos & Griebeler).
- κ (norm weight): intergenerational norm proxy residualized on current income.
- ϕ (law bite): DiD / event study around March 2019.

Three-Layer Empirical Strategy

- 1 **Descriptive / structural layer:** Estimate marriage-market model parameters using pre-2019 equilibrium; establish baseline mechanism shares.
- 2 **Mechanism quantification:** Heterogeneous-effects regression of PBF on marriage, stratified by norm proxy.
- 3 **Causal ID of Lei 13.811/2019:** DiD / event study / bunching analysis on formal and informal margins.

Guiding principle

The law's **total** causal effect (Layer 3) is only interpretable in light of the **substitution margins** made visible by Layers 1–2. Without the mechanism decomposition, a null or small law effect is ambiguous between “law doesn’t bind” and “law displaces to informal.”

Layer 1: Structural Estimation of Mechanism Weights

Estimate discrete-time hazard of marriage (formal, informal, wait) using [PNADC 2012–2019](#):

$$P(M_{it} = j \mid a_{it}, Y_{it}, N_i, X_{it}) = \frac{\exp(V_{it}^j)}{\sum_k \exp(V_{it}^k)}$$

with V^j from the utility specification. Yields:

- $\hat{\gamma}$: income elasticity of union formation by age.
- $\hat{\kappa}$: norm weight in union formation.
- $\hat{\mu}$: reputational sensitivity.
- **Counterfactual simulation**: what share of unions would occur if Y were raised to poverty line?
→ “norm-irreducible” share.

Layer 2: Mechanism Quantification via Heterogeneous Effects

Triple-interaction specification:

$$P(\text{Union}_i) = \alpha_m + \beta_1 \text{PBF}_i + \beta_2 \text{MomEarly}_i + \beta_3 (\text{PBF}_i \times \text{MomEarly}_i) + \gamma Y_i + X_i' \delta + \varepsilon_i$$

where

- $\text{MomEarly}_i = \mathbb{1}[\text{age at first union of mother} < 16]$ (PNADC/PNS);
- fallback proxy: mother's age – oldest child's age < 17 (Census);
- Y_i absorbs current income so MomEarly_i captures **norm persistence, not poverty persistence**.

Predictions:

- $\beta_1 < 0$: replicates PBF main effect.
- $\beta_2 > 0$: norm transmission.
- $\beta_3 > 0$: PBF bite is smaller in high-norm families → estimate of “norm-irreducible” share as $|\beta_1 + \beta_3|/|\beta_1|$.

Layer 2 (cont.): A Richer Norm Index

Single proxies carry measurement error. Build a composite index $\hat{N}_i$:

- Mother's age at first union < 16 (PNADC/PNS)
- Any older sister married before 18 (household roster)
- Share of women 25+ in census tract married before 18 (Census 2010)
- Religious affiliation (Census / PNS)
- Indigenous / quilombola self-identification (Census)
- Rural North/Northeast residence (IBGE region)

Principal-components or standardized-sum aggregation. Use PCA weights in β_3 interaction for robustness.

Three complementary designs:

- 1 **Event study on formal marriages** (Registro Civil, SIDRA/IBGE):

$$y_{at} = \alpha_a + \lambda_t + \sum_{k \neq -1} \theta_k \mathbb{1}[t - 2019 = k] \cdot \mathbb{1}[a < 16] + \varepsilon_{at}$$

Tests whether the formal-marriage hazard for under-16 girls drops at 2019.

- 2 **Age-based DiD**: under-16 (treated) vs. 16–17 (control) $\times$ pre/post 2019, across data sources (formal and informal).
- 3 **Bunching / McCrary test** (Collin 2016 methodology): discontinuity in the age-at-first-union density at 16, pre vs. post 2019.

Layer 3 (cont.): The Formal–Informal Comparison

Critical diagnostic for law effect:

Scenario	Formal (Reg. Civil)	Informal (PNADC/Census)	Interpretation
Law binds, no subst.	↓↓	↓	Real reduction
Law displaces margin	↓↓	↑ or ≈	Substitution only
Law doesn't bind	≈	≈	No enforcement

- BLANK ET AL. (2020) warning: admin data will overstate reduction.
- Triangulation with PNADC / PNS / Census informal-union measures is **mandatory**.
- SINASC (birth to under-16 mothers with “union” status) offers a cross-check.

Layer 3 (cont.): Heterogeneous Law Effects

Interact the law treatment with the norm index $\hat{N}_i$ and region:

$$y_{irt} = \alpha_r + \lambda_t + \theta_1 \text{Post}_t \cdot \text{Under16}_i + \theta_2 \text{Post}_t \cdot \text{Under16}_i \cdot \hat{N}_i + X_{irt} \beta + \varepsilon_{irt}$$

Predictions from the model:

- $\theta_1 < 0$: formal bite.
- $\theta_2 > 0$: **law works less well in high-norm regions/families** — the same “enforcement gradient” the global literature documents for LMIC bans (Collin 2016, Wilson 2022).

Policy implication: Lei 13.811/2019 alone cannot eliminate the residual; complementary interventions (à la Chow & Vivaldi 2022) needed in high-norm contexts.

Primary Data Sources

Source	Coverage	Use
PNADC (IBGE)	2012–present, quarterly	Informal + formal unions, age at first union, income, PBF status
PNS (IBGE)	2013, 2019	Prevalence, health, norms; pre/post snapshot of law
CENSO 2010 / CENSO 2022	Universal	Spatial heterogeneity, tract-level norm index, pre/post comparison
REGISTRO CIVIL (SIDRA/IBGE)	Annual, full coverage	Formal marriages by age and sex; primary DiD outcome
SINASC (DATA-SUS)	Annual	Births to under-16 mothers, union status at birth
SINAN (DATASUS)	Annual (2014–)	Sexual violence, marital rape; secondary outcomes
CADÚNICO	Monthly administrative	PBF eligibility, restricted access via MDS convenio

Key Variables

- **Outcome:** indicator of union (formal, informal, any) by age-year.
- **Treatment:** age $\times$ calendar year, with cutoff at Lei 13.811/2019 (March 2019).
- **Income:** household per-capita income; PBF receipt indicator; value received.
- **Norm index $\hat{N}_i$:** standardized composite — mother's age at first union, sister unions, tract-level early-union share, religion, self-declared race/ethnicity, region.
- **Controls:** household composition, mother's/father's education, race, region, urban/rural, municipal-level GDP and HDI.
- **Mechanism probes:** pregnancy at time of union, partner age gap, schooling status at union.

Next Steps: Immediate (0–3 months)

- 1 **Data pipeline:** pull PNADC 2012–2024, Registro Civil series (SIDRA), Census 2010, SINASC 2011–2022. Agree on variable naming and panel construction.
- 2 **Descriptive replication:** reproduce Cardoso (2022), Margarido (2024), and Vasconcelos & Griebeler (2023) baseline numbers. This validates our cleaning.
- 3 **Stylized-fact figures:** age-specific marriage hazard by year (2010–2024), pre/post 2019 visual.
- 4 **Norm index v1:** construct and validate $\hat{N}_i$ on PNADC.
- 5 **Pre-analysis plan draft:** register specifications for Layer 2 and Layer 3 before running regressions on full panel.

Next Steps: Medium (3–9 months)

- 1 **Structural estimation** (Layer 1): discrete-choice hazard model on pre-2019 PNADC. Compute “norm-irreducible” counterfactual.
- 2 **Heterogeneous PBF effects** (Layer 2): triple-interaction specification; publish the β_3 estimate as the central mechanism parameter.
- 3 **Law evaluation** (Layer 3):
 - Event study on Registro Civil.
 - Age-based DiD on PNADC (informal margin).
 - McCrary test for bunching pre/post 2019.
- 4 **Robustness**: alternative norm proxies; placebo cohorts; spatial heterogeneity (North/Northeast vs. rest).
- 5 **CadÚnico data access**: file request with MDS for administrative PBF data if feasible.

Next Steps: Longer Horizon (9–18 months)

- 1 **Qualitative triangulation**: pair the quantitative findings with selected interviews or use of Taylor & Fonseca / Margarido ethnographic evidence to interpret residuals.
- 2 **Secondary outcomes**: SINAN marital-rape trends pre/post law; SINASC births-in-union trends — does formal-margin compliance translate to child-welfare gains?
- 3 **Policy simulation**: use the structural model to simulate complementary interventions (norm campaigns à la Chow & Vivalt) on top of the law.
- 4 **Working paper** submission target: late 2026.

Open Questions for Coauthor Discussion

- 1 Is the **formal vs. informal margin** comparison credible with available data, or do we need a primary survey component?
- 2 Do we commit to a **full structural model** (Layer 1) or prioritize reduced-form identification (Layers 2–3)?
- 3 Is intergenerational norm transmission the right primary proxy, or should we lead with **regional / community-tract** variation?
- 4 How do we handle the **COVID-19 break** (2020–2022) in the post-2019 window?
- 5 **Division of labor** — who leads data construction, structural estimation, reduced-form estimation, qualitative triangulation?
- 6 Are there additional Brazilian datasets (e.g., SAEB school records linked to PNADC) worth pursuing?

The DCDP: Full Specification

Let $a \in \{10, \dots, 17\}$ be the girl's age and $p \in \{0, 1\}$ pregnancy status. Each period the household chooses $M \in \{F, I, W\}$. A match offer $(\Delta, w(\Delta))$ is drawn from the age-specific market distribution $F_\Delta(a)$.

$$U_i^F = \underbrace{w(\Delta)}_{\text{wealth transfer}} + \underbrace{\kappa_i N_i(\Delta)}_{\text{norm value}} + \underbrace{\mu_i R(a)}_{\text{reputational relief}} - \underbrace{\phi^F(a, p, t)}_{\text{legal cost}} + \varepsilon^F$$

$$U_i^I = w(\Delta) + (\kappa_i - \lambda) N_i(\Delta) + \mu_i R(a) - \phi^I(a) + \varepsilon^I$$

$$U_i^W = \underbrace{Y_i}_{\text{income}} + \underbrace{s(a)}_{\text{schooling}} - \underbrace{\sigma p}_{\text{pregnancy cost}} + \delta \mathbb{E}_{p'}[V(a+1, p')] + \varepsilon^W$$

With $\varepsilon \stackrel{\text{iid}}{\sim}$ Type-I EV, the optimal value satisfies $V(a, p) = \mathbb{E}_\Delta[\log(\exp V^F + \exp V^I + \exp V^W)]$ — computed via **backward induction** from $V(18, \cdot) \equiv 0$.

Legal Cost Functions: ϕ^F and ϕ^I

Formal margin (ϕ^F):

$$\phi^F(a, p, t) = \bar{\phi}^F \cdot 1[a < 16] \\ \cdot [1 - (1 - 1_{t \geq 2019}) \cdot p]$$

- Pre-2019: cost waived if pregnant ($p = 1$).
- Post-2019: **no exceptions** — $\bar{\phi}^F$ always active below 16.
- Identified from: DiD on Registro Civil around March 2019.

Informal margin (ϕ^I):

$$\phi^I(a) = \bar{\phi}^I \cdot 1[a < 16]$$

- Reporting risk if union made known to authorities.
- **Unaffected by Lei 13.811/2019.**
- Identified from: PNADC informal-union rates at age threshold.

Substitution prediction

If $\bar{\phi}^I \ll \bar{\phi}^F$, the 2019 reform shifts unions from formal to informal rather than reducing total child marriage.

Two-Stage Estimation Strategy

Stage 1 — Registro Civil DiD (reduced form)

Estimate $\hat{\phi}^F$ from the aggregate formal-marriage panel:

$$\log n_{a,s,t} = \alpha_s + \alpha_a + \alpha_t + \underbrace{\theta \cdot 1[a < 16] \cdot 1[t \geq 2019]}_{\hat{\phi}^F} + \varepsilon_{a,s,t}$$

State (s), age-group (a), year (t) fixed effects; SEs clustered by state. Data: Registro Civil 2003–2022 (IBGE/SIDRA).

Stage 2 — Structural MLE (PNAD-C)

Given $\hat{\phi}^F$, maximize weighted log-likelihood over $\Theta = (\alpha_0, \alpha_1, \mu, \gamma_0, \gamma_1, \beta_0, \beta_1, \sigma)$:

$$\hat{\Theta} = \arg \max_{\Theta} \sum_i w_i \left[y_i \log P_i^{\text{union}}(\Theta) + (1 - y_i) \log(1 - P_i^{\text{union}}(\Theta)) \right]$$

where $P_i^{\text{union}} = \text{logistic}(V_i^{\text{union}} - V_i^{\text{wait}})$ and V_i^{union} integrates over the empirical $F_{\Lambda}(a)$ via Monte Carlo

PNAD-C sample (girls 10–17)

Variable	Mean (SD)	
	In union	Wait
<i>N</i>	4,671	821
Age <i>a</i>	16.5	15.8
Age gap Δ	8.4 (6.3)	—
Partner income (R\$)	1,014	—
Attending school	30%	59%
Parda/preta	87%	84%
Rural	28%	24%
Post-2019	15%	19%

Registro Civil panel (minor brides)

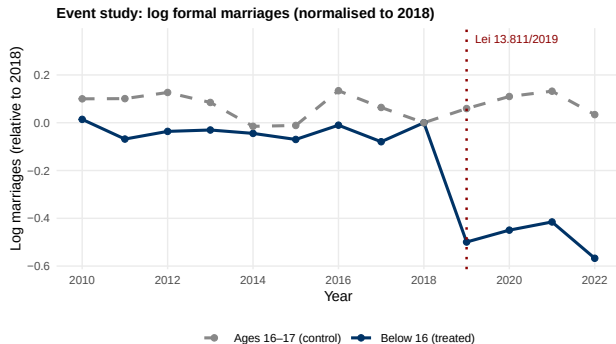
Variable	Value
Years	2003–2022
States	27
Age groups	5
Below-16 cell	2
Total cells	2,970
<i>Below-16 formal marriages:</i>	
Mean (pre-2019)	37.2/state/yr
Mean (post-2019)	35.0/state/yr
Δ post-reform	–5.9%

Note: School attendance and race pooled over PNAD-C 2012–2023; survey weights applied throughout.

Stage 1 Results: DiD on Formal Marriages

	Dep. var.: $\log(n_{ast} + 1)$	
	(1)	(2)
$1[a < 16] \times 1[t \geq 2019]$	0.411*** (0.022)	0.398*** (0.025)
State FE	✓	✓
Age-group FE	✓	✓
Year FE	✓	✓
State trends		✓
N	2,970	2,970
R^2	0.89	0.91

Note: SEs clustered by state. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.



Interpretation

The positive coefficient implies formal marriages *increased* for under-16 girls post-2019 relative to the 16-17 control group — consistent with a **substitution reversal**: families may have rushed to formal registration before

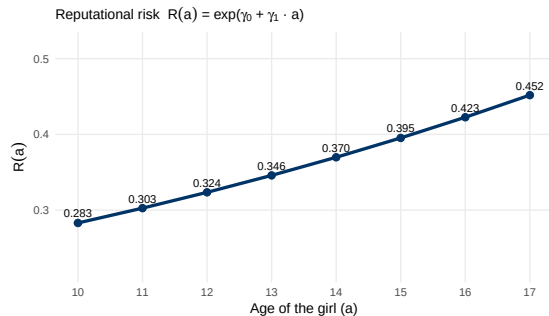
Stage 2 Results: Structural Parameter Estimates

	Parameter	Estimate	Boot. SE	95% CI	<i>t</i>
<i>Economic mechanism</i>					
α_0	Union entry (intercept)	−0.642	0.213	[−1.13, −0.18]	−3.0**
α_1	$w(\Delta)$ slope	−0.009	0.007	[−0.016, 0.010]	−1.3
<i>Norm and reputation mechanism</i>					
μ	Reputation weight	1.067	0.091	[0.89, 1.23]	0.76 [†]
γ_0	$\log R(a)$ intercept	−1.934	0.085	[−2.14, −1.83]	−22.7***
γ_1	$\log R(a)$ age slope	0.067	0.013	[0.047, 0.097]	5.1***
<i>Schooling and outside option</i>					
β_0	$s(a)$ intercept	−3.245	1.610	[−6.34, −0.34]	−2.0*
β_1	$s(a)$ slope in a	0.054	0.105	[−0.132, 0.257]	0.5
σ	Pregnancy cost	<i>unidentified</i> (see text)			—
<i>First-stage (Registro Civil DiD)</i>					
$\hat{\phi}^F$	Legal deterrence cost	0.411	0.022		18.7***

[†]log-scale *t*-stat; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. $N = 5,492$. $B = 200$ bootstrap replications.

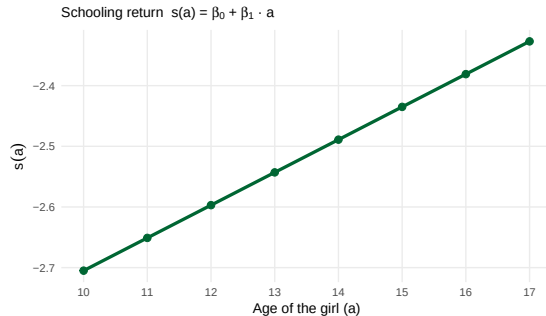
Log-likelihood: −2,487.8. Discount factor $\delta = 0.95$ calibrated. κ fixed at 0 (collinear with α_0).

Derived Economic Quantities



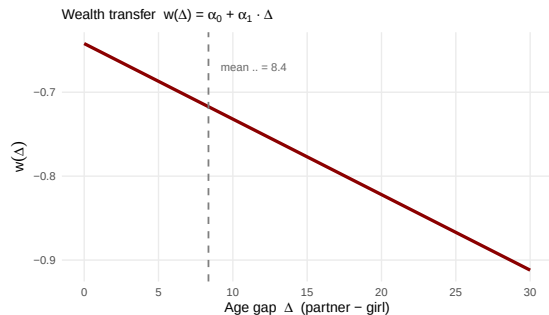
$R(a)$ rises **59%** from age 10 (0.28) to 17 (0.45).

Together with $\hat{\mu} = 1.07$, this implies the
norm-reputational payoff to marriage grows substantially with age.



$s(a)$ rises mildly from -2.70 to -2.32 . The negative level means the outside-option schooling return is dominated by the union payoff in the data — consistent with the **85% in-union share.**

Wealth Transfer $w(\Delta)$ and Mechanism Decomposition



Estimated $\hat{\alpha}_1 = -0.009$ (n.s.): the age-gap slope is essentially flat in the current specification. This suggests Δ alone does not proxy well for $w(\Delta)$ — partner income (Y_{parc}) should enter directly in the next iteration.

Mechanism	Share of V^{union}
$w(\Delta)$ at avg. gap	-0.71
$\mu R(a)$ at age 10	+0.30
$\mu R(a)$ at age 17	+0.48
Net V^{union} , age 10	-0.41
Net V^{union} , age 17	-0.23
V^{wait} , age 10	-2.70
V^{wait} , age 17	-2.32

Key finding

The union option dominates waiting at all ages. The **reputational mechanism** ($\mu R(a)$) accounts for nearly all of the age gradient in marriage propensity.

Identification Concerns and Model Limitations

What is identified

- $\hat{\phi}^F$: clean DiD off the 2019 reform.
- $R(a)$ shape: age gradient in union propensity (conditional on income and Δ).
- $s(a)$: outside-option level via school-attendance variation.
- μ : scale of reputational channel.

What is not yet identified

- σ (pregnancy cost): `matern_bin` is near-zero in PNAD-C; calibrate from SINASC.
- λ (formal vs. informal disutility): needs formal/informal split in PNAD-C.
- $\bar{\phi}'$ (informal legal cost): needs enforcement variation.
- κ_i heterogeneity: fixed $\kappa = 0$; norm index enters in Layer 2 regression.

Roadmap to full identification

- 1 Calibrate σ from SINASC birth-in-union rates by age.
- 2 Recover λ from PNAD-C union-type variable (`V2005`).
- 3 Interact $\hat{\kappa}_i$ (norm index) with PNAD-C to estimate Layer 2 triple interaction β_3 .

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Thank you

Discussion and feedback welcome

Working document — not for circulation outside the author team